DO NOW — RECALL

Lesson 144 · DNS and IP Routing: From Name to Destina

Name: _____ Class: _____ Date: _____

1. Which TCP/IP layer handles routing and IP addressing?

2. What data structure uses LIFO?

3. What is a trace table used for?

4. How many bits are in one byte?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Internet layer.

2. Model answer

Stack.

3. Model answer

Tracking variable state changes as an algorithm runs.

4. Model answer

Eight.

By the end, you can…

• Browser requests address for a website name through DNS.

• Application connects to a resolved IP address and transport port.

• Routers forward packets hop by hop to the destination network.

Vocabulary — know these terms

DNS — Hierarchical name-resolution system.

IP address — Network-layer address used to route packets.

resolver — Client/service that queries DNS and caches answers.

router — Forwards IP packets towards destination networks.

NEW LEARNING

Read, annotate and explain

Resolve hostname

Browser requests address for a website name through DNS.

Use IP destination

Application connects to a resolved IP address and transport port.

Route across networks

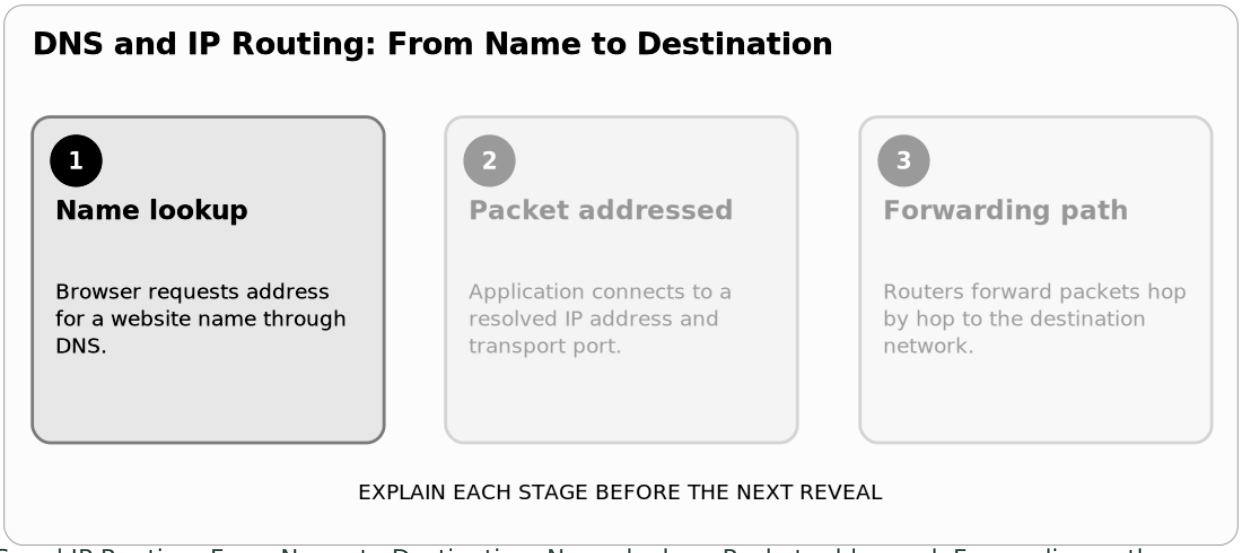
Routers forward packets hop by hop to the destination network.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

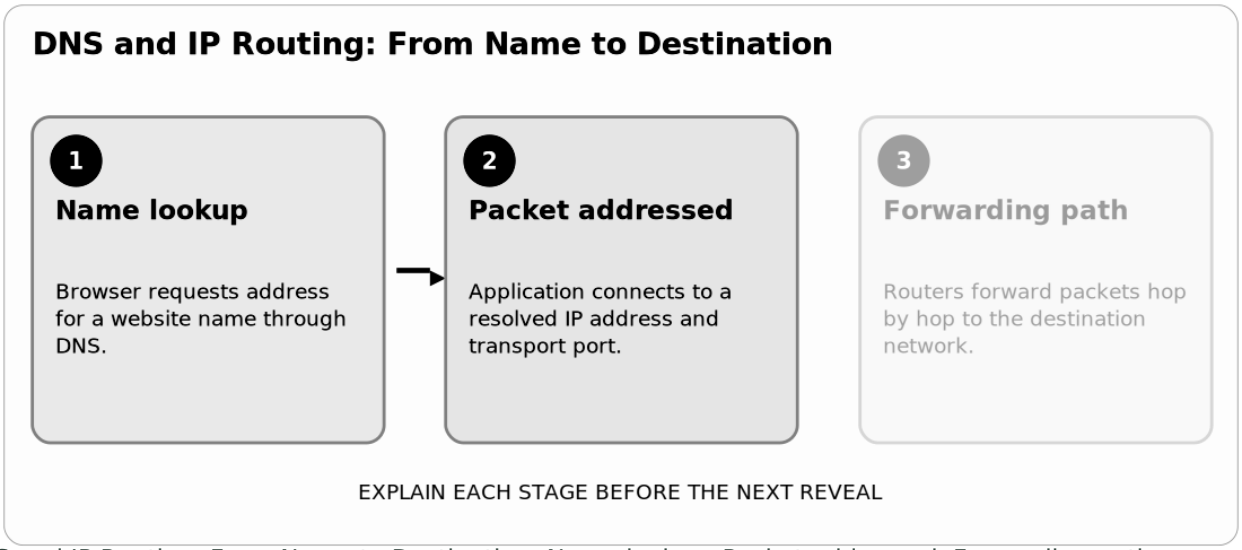
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



DNS and IP Routing: From Name to Destination: Name lookup, Packet addressed, Forwarding path

New Learning

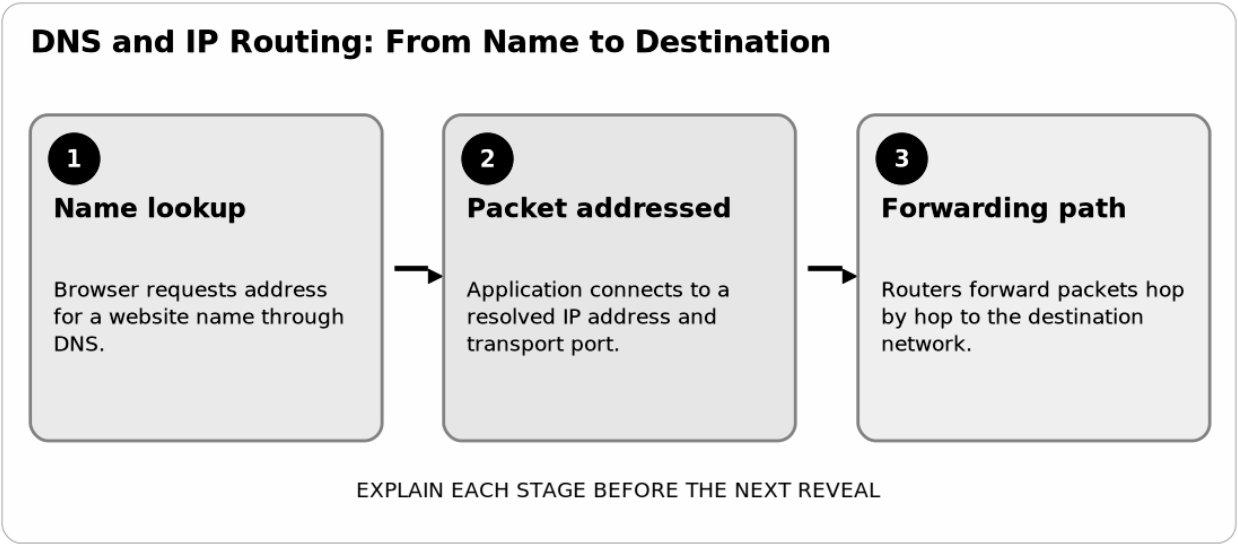


DNS and IP Routing: From Name to Destination: Name lookup, Packet addressed, Forwarding path

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



DNS and IP Routing: From Name to Destination: Name lookup, Packet addressed, Forwarding path

I DO — WORKED EXAMPLE

Follow the worked method

A pupil types example.org and receives an IP address from DNS. Explain what DNS does and does not do.

Name example.org

Identify the relevant information and make a prediction.

DNS returns an IP address

Explain the change using correct technical terms.

Connect to web server

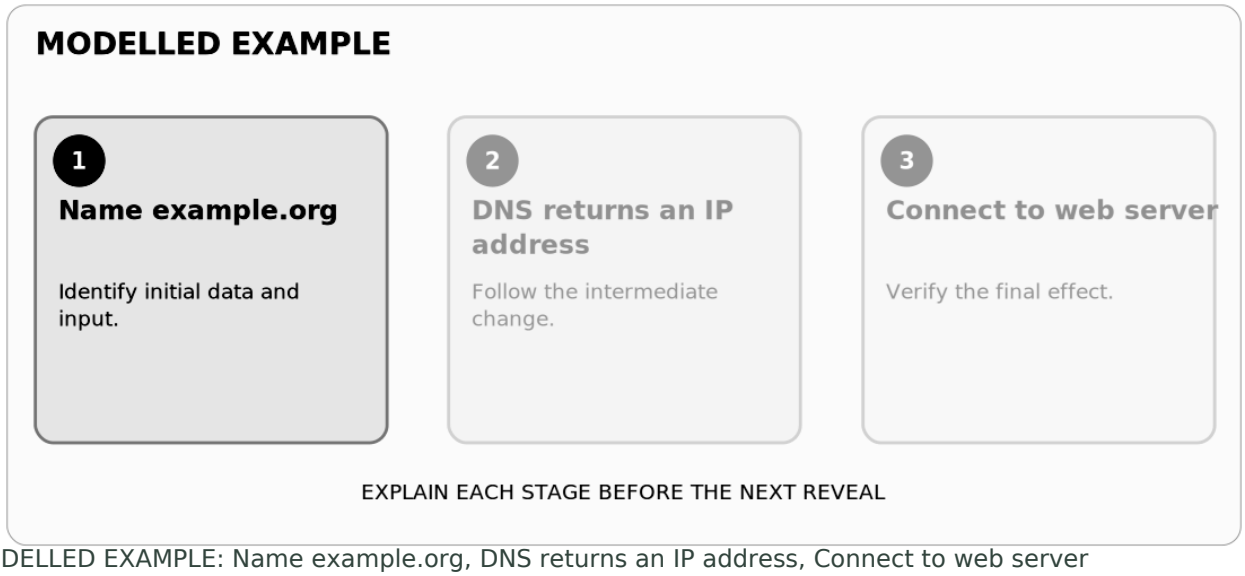
DNS maps the hostname to relevant resource records such as an A/AAAA address. DNS does not directly transmit the web page. Browser makes a separate connection to the chosen server address and requests content, often over HTTPS.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A pupil types example.org and receives an IP address from DNS. Explain what DNS does and does not do.

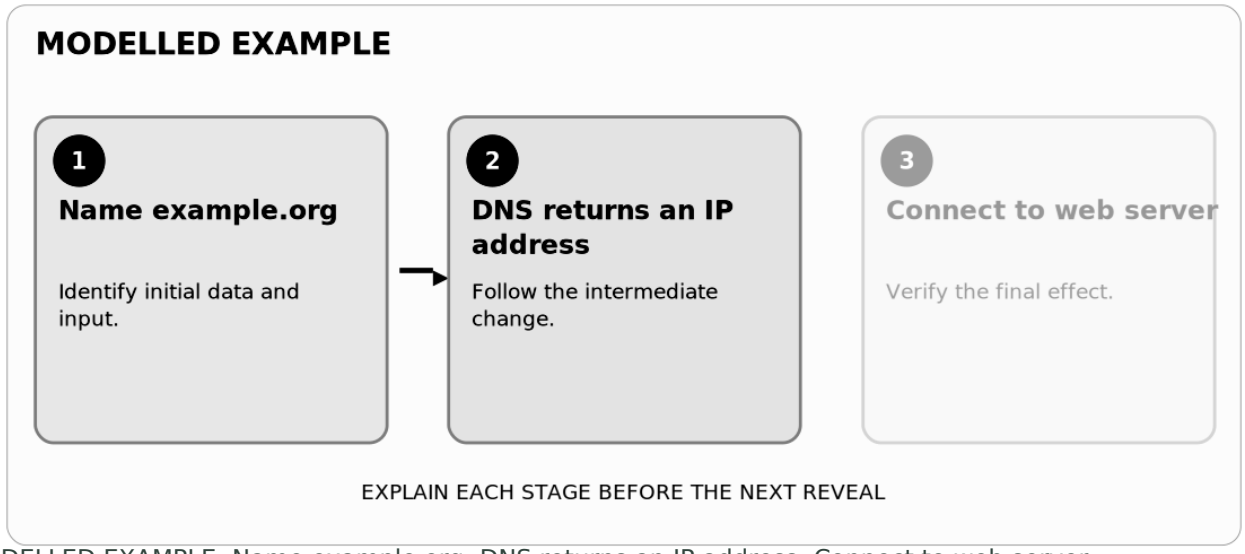
Identify the relevant information and make a prediction.



MODELLED EXAMPLE: Name example.org, DNS returns an IP address, Connect to web server

A pupil types example.org and receives an IP address from DNS. Explain what DNS does and does not do.

Identify the relevant information and make a prediction.



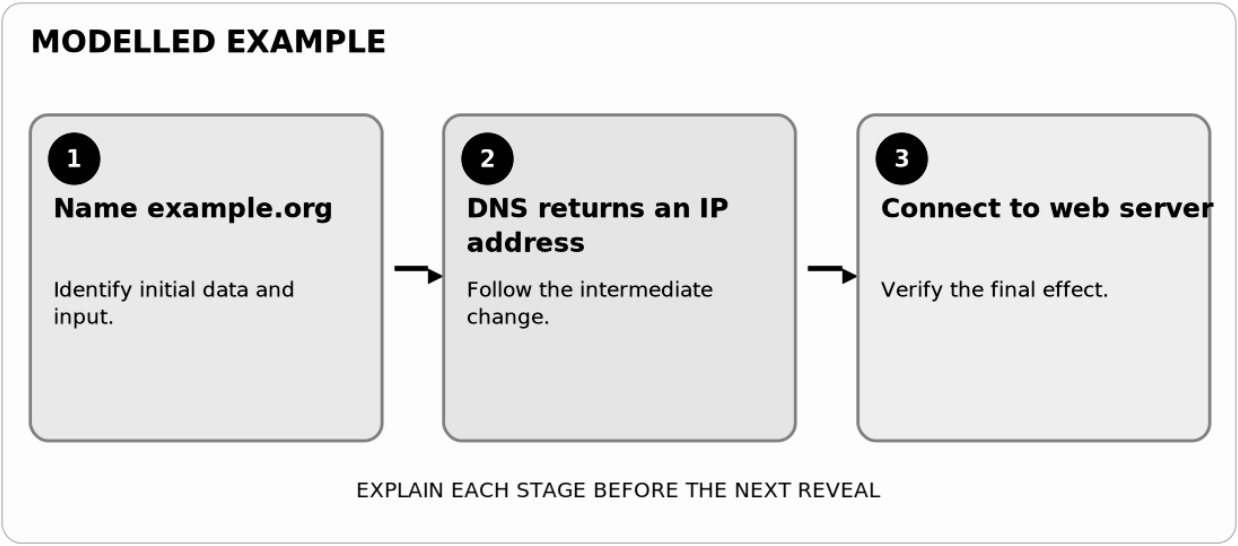
MODELLED EXAMPLE: Name example.org, DNS returns an IP address, Connect to web server

I Do · original lesson diagrams

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A pupil types example.org and receives an IP address from DNS. Explain what DNS does and does not do.

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MODELLED EXAMPLE: Name example.org, DNS returns an IP address, Connect to web server

WE DO — GUIDED PRACTICE

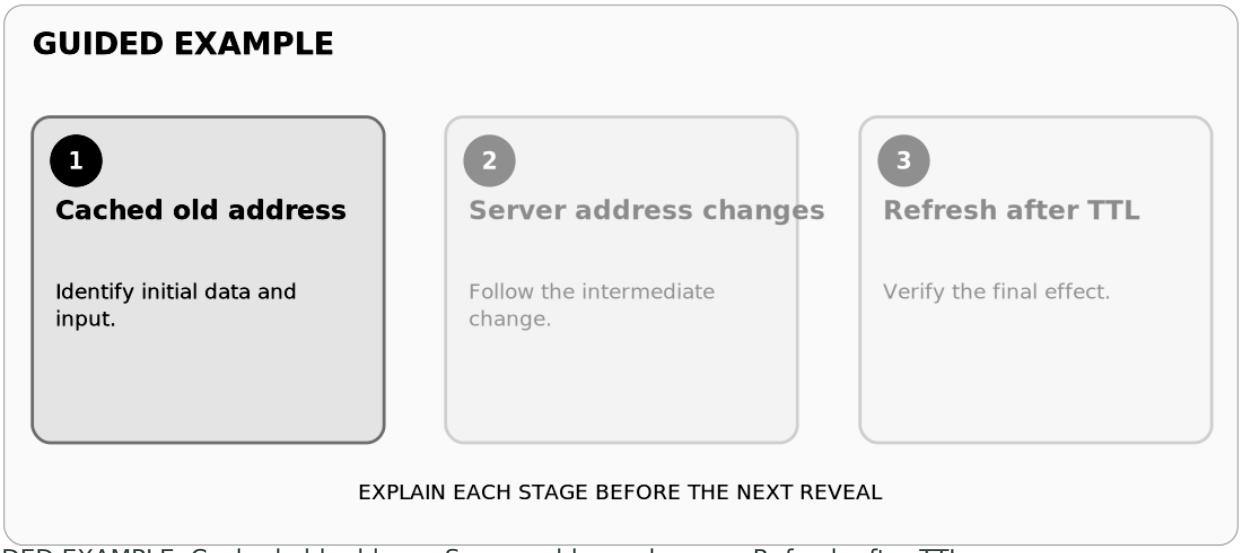
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A DNS answer has been cached but website server has moved. Explain a possible issue and resolution.

Identify the relevant information and make a prediction.



GUIDED EXAMPLE: Cached old address, Server address changes, Refresh after TTL

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Cached old address Identify the relevant information and make a prediction.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain how DNS helps a user retrieve a website by name. [6]

Question 2

Differentiate DNS name resolution from IP packet routing. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A school website moves servers but some users reach the old address. Explain cached DNS and propose sensible checks. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

DNS sends the entire web page to the browser after it receives the hostname.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

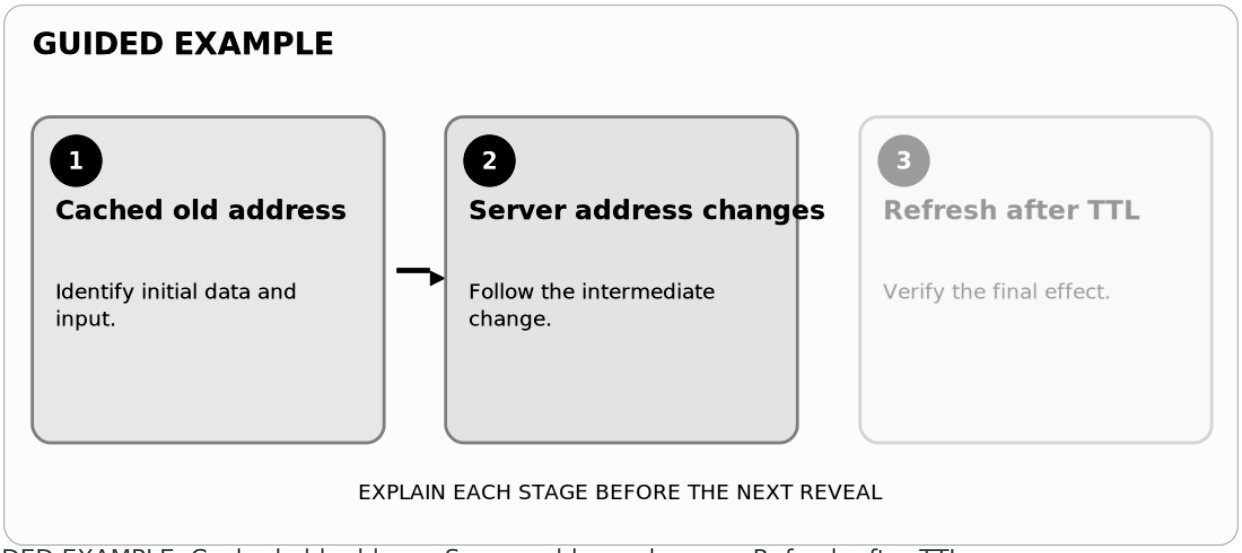
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A DNS answer has been cached but website server has moved. Explain a possible issue and resolution.

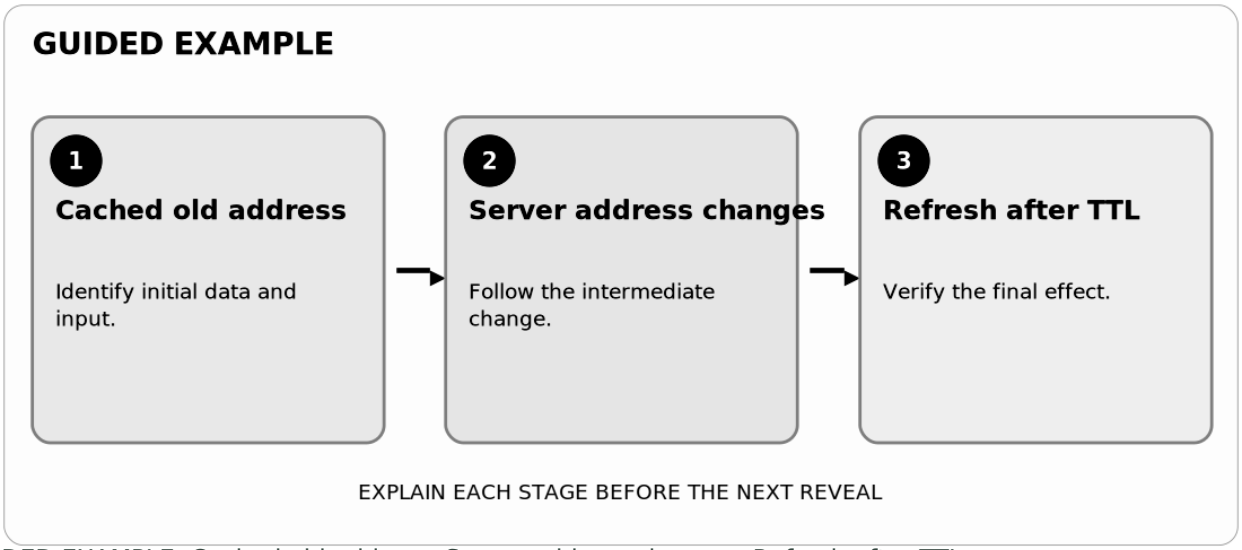
Identify the relevant information and make a prediction.



GUIDED EXAMPLE: Cached old address, Server address changes, Refresh after TTL

A DNS answer has been cached but website server has moved. Explain a possible issue and resolution.

Identify the relevant information and make a prediction.



GUIDED EXAMPLE: Cached old address, Server address changes, Refresh after TTL

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Server address changes

Explain the change using correct technical terms.

Refresh after TTL

The cached address may remain until its TTL expires; client may contact an outdated endpoint. The resolver can refresh after expiry. DNS propagation and authoritative changes do not guarantee immediate worldwide consistency.

IMPROVE: Explain one correction to your own reasoning.